B. Synteny-Anchored Locus Retrieval Pipeline Across 68 Genomes

Identification of homologous intervals via conserved syntenic anchors and discovery of 4 macroevolutionary archetypes

Synteny Discovery Pipeline & Chromosome Contiguity Filtering

1. Syntenic Search

Query 68 genomes for flanking markers:
α: NPRL3, MPG, RHBDF1, LUC7L
β: OR51 cluster, CDCA8, TRIM21, RRM1

2. Interval Bounds

Define syntenic window containing all flankers and globin duplications.
Extract verified coding models.

3. Scaffold vs Chr

Discriminate assembled chromosomes from unplaced scaffolds to avoid artificial telomeric misassignment.

4. Baseline Calibration

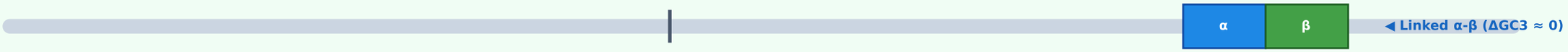
Compute whole-genome CDS & gene baselines (mean, SD, percentiles) to calculate locus z-score deviations.

1. Linked Primordial Archetype (Basal Actinopterygii & Amphibians — Spotted Gar, Xenopus)

Ancestral tandem arrangement: α and β globin genes reside on the same chromosome in a single cluster.

Both genes share identical genomic environment and regional recombination rate. ΔGC3 (α - β) ≈ 0%.

Cytogenetic State: Single Chromosome (Gar Chr 19 / Xenopus Chr 1)



2. Canonical Decoupled Subtelomeric-α (Eutherian & Metatherian Mammals — Human, Mouse)

Amniote translocation decouples α and β. α captured by subtelomere (<1 Mb to telomere); β remains interstitial.

Runaway subtelomeric gBGC supercharges α-GC3 (>85–95%) while β-GC3 remains moderate (~60–65%). ΔGC3 >> 0%.

Cytogenetic State: Two Chromosomes (α Subtelomeric / β Interstitial)

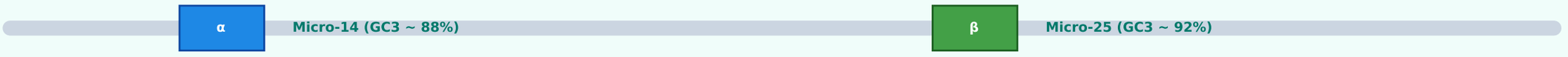


3. Dual Microchromosomal Partitioning (Archosauria & Aves — Chicken, Eagle, Hummingbird)

Both α and β clusters translocate onto distinct, highly recombinogenic microchromosomes (Chr 14 and Chr 25).

High crossover density drives synchronous hyper-elevation of both clusters: α-GC3 > 85%, β-GC3 > 85%. ΔGC3 ≈ 0%.

Cytogenetic State: Two Microchromosomes (Micro-14 and Micro-25)



4. The Great Inversion (Squamate Reptiles — Lizards & Snakes — Wall Lizard, Garter Snake)

Karyotypic inversion translocates β-locus to microchromosome/subtelomere, leaving α in an interstitial desert.

Inverted gBGC polarity: β-GC3 skyrockets (>90–96%) while α-GC3 languishes (~48–55%). ΔGC3 << 0% (β >> α)!

Cytogenetic State: Inverted Polarity (β Subtelomeric / α Interstitial)

